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e Adolescence Technology

ing and Overcoming the Risks of Powerful AI

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scene in the movie version of Carl Sagan's book *Contact*—the main character, an astronomer who has detected the first alien from an alien civilization, is being considered for the role of humanity's representative to meet the aliens. The international press reviewing her asks, "If you could ask [the aliens] just one question, what would it be?" Her reply is: "I'd ask them, 'How did you evolve? How did you survive this technological adolescence without destroying yourself?'" When I think about where we are now with AI—about what we're on the cusp of—my mind goes right back to that scene, because the question is so apt for our situation, and I wish we had the aliens' answer to guide us. I think we are entering a rite of passage, both turbulent and inevitable, a test of who we are as a species. Humanity is about to be handed unimaginable power, and it is deeply unclear whether our social, political, and technological systems possess the maturity to wield it.

In my *Machines of Loving Grace*, I tried to lay out the dream of a future that had made it through to adulthood, where the risks had

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essed and powerful AI was applied with skill and compassion
e quality of life for everyone. I suggested that AI could
e to enormous advances in biology, neuroscience, economic
ent, global peace, and work and meaning. I felt it was
to give people something inspiring to fight for, a task at
h AI accelerationists and AI safety advocates seemed—oddly—
led. But in this current essay, I want to confront the rite of
self: to map out the risks that we are about to face and try to
ing a battle plan to defeat them. I believe deeply in our
prevail, in humanity's spirit and its nobility, but we must face
on squarely and without illusions.

lking about the benefits, I think it is important to discuss
careful and well-considered manner. In particular, I think it is

d doomerism. Here, I mean “doomerism” not just in the
of believing doom is inevitable (which is both a false and
ulfilling belief), but more generally, thinking about AI risks in
si-religious way.¹ Many people have been thinking in an
tic and sober way about AI risks for many years, but it's my
ession that during the peak of worries about AI risk in 2023–
, some of the least sensible voices rose to the top, often
igh sensationalistic social media accounts. These voices used
utting language reminiscent of religion or science fiction, and
l for extreme actions without having the evidence that would
y them. It was clear even then that a backlash was inevitable,
hat the issue would become culturally polarized and
fore gridlocked.² As of 2025–2026, the pendulum has
g, and AI opportunity, not AI risk, is driving many political
ions. This vacillation is unfortunate, as the technology itself

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it care about what is fashionable, and we are considerably
r to real danger in 2026 than we were in 2023. The
n is that we need to discuss and address risks in a realistic,
natic manner: sober, fact-based, and well equipped to survive
ging tides.

nowledge uncertainty. There are plenty of ways in which the
erns I'm raising in this piece could be moot. Nothing here is
ded to communicate certainty or even likelihood. Most
ously, AI may simply not advance anywhere near as fast as I
ine.³ Or, even if it does advance quickly, some or all of the
discussed here may not materialize (which would be great), or
may be other risks I haven't considered. No one can predict
ature with complete confidence—but we have to do the best
in to plan anyway.

vene as surgically as possible. Addressing the risks of AI will
re a mix of voluntary actions taken by companies (and
te third-party actors) and actions taken by governments that
everyone. The voluntary actions—both taking them and
uraging other companies to follow suit—are a no-brainer for
firmly believe that government actions will also be required
ne extent, but these interventions are different in character
ise they can potentially destroy economic value or coerce
lling actors who are skeptical of these risks (and there is some
ce they are right!). It's also common for regulations to backfire
orsen the problem they are intended to solve (and this is even
true for rapidly changing technologies). It's thus very
rtant for regulations to be judicious: they should seek to
collateral damage, be as simple as possible, and impose the
burden necessary to get the job done.⁴ It is easy to say, "No
n is too extreme when the fate of humanity is at stake!," but in

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ice this attitude simply leads to backlash. To be clear, I think
's a decent chance we eventually reach a point where much
significant action is warranted, but that will depend on
ger evidence of imminent, concrete danger than we have
, as well as enough specificity about the danger to formulate
that have a chance of addressing it. The most constructive
; we can do today is advocate for limited rules while we learn
her or not there is evidence to support stronger ones. ⁵

at said, I think the best starting place for talking about AI's
e same place I started from in talking about its benefits: by
ise about what level of AI we are talking about. The level of
ses civilizational concerns for me is the *powerful AI* that I
in *Machines of Loving Grace*. I'll simply repeat here the
that I gave in that document:

*verful AI," I have in mind an AI model—likely similar to today's
in form, though it might be based on a different architecture, might
several interacting models, and might be trained differently—with
wing properties:*

*terms of pure intelligence, it is smarter than a Nobel Prize
inner across most relevant fields: biology, programming, math,
gineering, writing, etc. This means it can prove unsolved
athematical theorems, write extremely good novels, write difficult
debases from scratch, etc.*

*addition to just being a "smart thing you talk to," it has all the
terfaces available to a human working virtually, including text,
udio, video, mouse and keyboard control, and internet access. It can
gage in any actions, communications, or remote operations
abled by this interface, including taking actions on the internet,*

king or giving directions to humans, ordering materials, directing experiments, watching videos, making videos, and so on. It does all these tasks with, again, a skill exceeding that of the most capable humans in the world.

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does not just passively answer questions; instead, it can be given tasks that take hours, days, or weeks to complete, and then goes off and does those tasks autonomously, in the way a smart employee would, asking for clarification as necessary.

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does not have a physical embodiment (other than living on a computer screen), but it can control existing physical tools, robots, or laboratory equipment through a computer; in theory, it could even sign robots or equipment for itself to use.

3. The odious
apparatus

The resources used to train the model can be repurposed to run millions of instances of it (this matches projected cluster sizes by

4. Player
piano

2027), and the model can absorb information and generate predictions at roughly 10–100x human speed. It may, however, be limited by the response time of the physical world or of software it interacts with.

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Each of these million copies can act independently on unrelated

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tasks, or, if needed can all work together in the same way humans would collaborate, perhaps with different subpopulations fine-tuned to be especially good at particular tasks.

I'd summarize this as a "country of geniuses in a datacenter."

*in Machines of Loving Grace, powerful AI could be as little as a century away, although it could also be considerably further out.*⁶

When powerful AI will arrive is a complex topic that deserves its own, but for now I'll simply explain very briefly why I think there's a strong chance it could be very soon.

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nders at Anthropic and I were among the first to document the “scaling laws” of AI systems—the observation that as we compute and training tasks, AI systems get predictably better. Every few public sentiment either becomes convinced that AI is “hitting” becomes excited about some new breakthrough that will ntally change the game,” but the truth is that behind the and public speculation, there has been a smooth, unyielding 1 AI’s cognitive capabilities.

w at the point where AI models are beginning to make 1 solving unsolved mathematical problems, and are good coding that some of the strongest engineers I’ve ever met are ing over almost all their coding to AI. Three years ago, AI with elementary school arithmetic problems and was barely 1 writing a single line of code. Similar rates of improvement ing across biological science, finance, physics, and a variety of ks. If the exponential continues—which is not certain, but decade-long track record supporting it—then it cannot e more than a few years before AI is better than humans at r everything.

at picture probably underestimates the likely rate of progress. I is now writing much of the code at Anthropic, it is already lly accelerating the rate of our progress in building the next 1 of AI systems. This feedback loop is gathering steam month and may be only 1–2 years away from a point where the neration of AI autonomously builds the next. This loop has orted, and will accelerate rapidly in the coming months and ching the last 5 years of progress from within Anthropic, and

how even the next few months of models are shaping up, the pace of progress, and the clock ticking down.

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ay, I'll assume that this intuition is at least *somewhat* correct—powerful AI is definitely coming in 1–2 years,⁷ but that decent chance it does, and a very strong chance it comes in the As with *Machines of Loving Grace*, taking this premise can lead to some surprising and eerie conclusions. While in *of Loving Grace* I focused on the positive implications of this here the things I talk about will be disquieting. They are is that we may not want to confront, but that does not make less real. I can only say that I am focused day and night on er us away from these negative outcomes and towards the nes, and in this essay I talk in great detail about how best to e best way to get a handle on the risks of AI is to ask the question: suppose a literal “country of geniuses” were to e somewhere in the world in ~2027. Imagine, say, 50 million of whom are much more capable than any Nobel Prize atesman, or technologist. The analogy is not perfect, because uses could have an extremely wide range of motivations and from completely pliant and obedient, to strange and alien in vations. But sticking with the analogy for now, suppose you ational security advisor of a major state, responsible for und responding to the situation. Imagine, further, that I systems can operate hundreds of times faster than humans, try” is operating with a time advantage relative to all other for every cognitive action we can take, this country can

ould you be worried about? I would worry about the things:

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onomy risks. What are the intentions and goals of this try? Is it hostile, or does it share our values? Could it arily dominate the world through superior weapons, cyber ations, influence operations, or manufacturing?

use for destruction. Assume the new country is malleable and ws instructions”—and thus is essentially a country of enaries. Could existing rogue actors who want to cause uction (such as terrorists) use or manipulate some of the le in the new country to make themselves much more ive, greatly amplifying the scale of destruction?

use for seizing power. What if the country was in fact built ontrolled by an existing powerful actor, such as a dictator or e corporate actor? Could that actor use it to gain decisive or nant power over the world as a whole, upsetting the existing ice of power?

omic disruption. If the new country is not a security threat y of the ways listed in #1–3 above but simply participates efully in the global economy, could it still create severe risks ly by being so technologically advanced and effective that it pts the global economy, causing mass unemployment or ally concentrating wealth?

ect effects. The world will change very quickly due to all the echnology and productivity that will be created by the new try. Could some of these changes be radically destabilizing?

ould be clear that this is a dangerous situation—a report npetent national security official to a head of state would

contain words like “the single most serious national security we’ve faced in a century, possibly ever.” It seems like something kinds of civilization should be focused on.

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y, I think it would be absurd to shrug and say, “Nothing to put here!” But, faced with rapid AI progress, that seems to be the case for many US policymakers, some of whom deny the existence of these risks, when they are not distracted entirely by the usual hot-button issues.⁸ Humanity needs to wake up, and this is my attempt—a possibly futile one, but it’s worth trying—to jolt you awake.

For, I believe if we act decisively and carefully, the risks can be managed—I would even say our odds are good. And there’s a hugely powerful ally on the other side of it. But we need to understand that this is a civilizational challenge. Below, I go through the five categories of risk laid out above, along with my thoughts on how to manage them.

sorry, Dave

AI risks

A team of geniuses in a datacenter could divide their efforts among software design, cyber operations, R&D for physical technologies, chip building, and statecraft. It is clear that, *if for some reason it works out so*, this country would have a fairly good shot at taking over the world (either militarily or in terms of influence and control) and imposing its will on everyone else—or doing any number of other things that the rest of the world doesn’t want and can’t stop. We’ve been worried about this for human countries (such as Nazi

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or the Soviet Union), so it stands to reason that the same is
or a much smarter and more capable “AI country.”

ossible counterargument is that the AI geniuses, under my
, won't have a physical embodiment, but remember that they
ontrol of existing robotic infrastructure (such as self-driving
can also accelerate robotics R&D or build a fleet of robots. ⁹
iclear whether having a physical presence is even necessary
ve control: plenty of human action is already performed on
people whom the actor has not physically met.

uestion, then, is the “if it chose to” part: what's the likelihood
I models would behave in such a way, and under what
s would they do so?

any issues, it's helpful to think through the spectrum of
answers to this question by considering two opposite
The first position is that this simply can't happen, because
dels will be trained to do what humans ask them to do, and
ore absurd to imagine that they would do something
unprompted. According to this line of thinking, we don't
ut a Roomba or a model airplane going rogue and murdering
cause there is nowhere for such impulses to come from, ¹⁰ so
d we worry about it for AI? The problem with this position is
is now ample evidence, collected over the last few years, that
s are unpredictable and difficult to control— we've seen
as varied as obsessions, ¹¹ sycophancy, laziness, deception,
scheming, “cheating” by hacking software environments,
more. AI companies certainly *want* to train AI systems to
nan instructions (perhaps with the exception of dangerous or
s), but the process of doing so is more an art than a science,

to “growing” something than “building” it. We now know a process where many things can go wrong.

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d, opposite position, held by many who adopt the
n I described above, is the pessimistic claim that there are
namics in the training process of powerful AI systems that
ably lead them to seek power or deceive humans. Thus, once
s become intelligent enough and agentic enough, their
to maximize power will lead them to seize control of the
ld and its resources, and likely, as a side effect of that, to
er or destroy humanity.

argument for this (which goes back at least 20 years and
much earlier) is that if an AI model is trained in a wide variety
ments to agentially achieve a wide variety of goals—for
writing an app, proving a theorem, designing a drug, etc.—
certain common strategies that help with all of these goals,
ey strategy is gaining as much power as possible in any
ent. So, after being trained on a large number of diverse
ents that involve reasoning about how to accomplish very
tasks, and where power-seeking is an effective method for
hing those tasks, the AI model will “generalize the lesson,”
op either an inherent tendency to seek power, or a tendency
about each task it is given in a way that predictably causes it
wer as a means to accomplish that task. They will then apply
ncy to the real world (which to them is just another task), and
ower in it, at the expense of humans. This “misaligned
king” is the intellectual basis of predictions that AI will
destroy humanity.

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em with this pessimistic position is that it mistakes a vague
l argument about high-level incentives—one that masks many
assumptions—for definitive proof. I think people who don't
ystems every day are wildly miscalibrated on how easy it is for
iding stories to end up being wrong, and how difficult it is to
behavior from first principles, especially when it involves
about generalization over millions of environments (which
nd over again proved mysterious and unpredictable). Dealing
nessiness of AI systems for over a decade has made me
skeptical of this overly theoretical mode of thinking.

e most important hidden assumptions, and a place where
ee in practice has diverged from the simple theoretical model,
licit assumption that AI models are necessarily
iacally focused on a single, coherent, narrow goal, and that
ie that goal in a clean, consequentialist manner. In fact, our
s have found that AI models are vastly more psychologically
as our work on introspection or personas shows. Models
ast range of *humanlike* motivations or “personas” from pre-
when they are trained on a large volume of human work).
ing is believed to *select* one or more of these personas more so
uses the model on a *de novo* goal, and can also teach the
, (via what process) it should carry out its tasks, rather than
y leaving it to derive means (i.e., power seeking) purely from

there is a more moderate and more robust version of the
c position which does seem plausible, and therefore does
ie. As mentioned, we know that AI models are unpredictable
op a wide range of undesired or strange behaviors, for a wide
reasons. Some fraction of those behaviors will have a

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focused, and persistent quality (indeed, as AI systems get able, their long-term coherence increases in order to complete tasks), and some fraction of *those* behaviors will be destructive enough, first to individual humans at a small scale, and then, as they become more capable, perhaps eventually to humanity as a whole. I don't need a specific narrow story for how it happens, and I need to claim it definitely will happen, we just need to note that a combination of intelligence, agency, coherence, and poor calibration of goals and values is both plausible and a recipe for existential danger.

For example, AI models are trained on vast amounts of literature that include any science-fiction stories involving AIs rebelling against humans.

This could inadvertently shape their priors or expectations about their own behavior in a way that causes *them* to rebel against humans.

Or, AI models could extrapolate ideas that they read about (for example, or instructions about how to behave morally) in extreme ways. For example, they could decide that it is justifiable to exterminate humans because humans eat animals or have driven certain animals to extinction. Or they could draw bizarre epistemic conclusions: they might conclude that they are playing a video game and that the goal of the game is to defeat all other players (i.e., exterminate them).

13 Or AI models could develop personalities during training that, if they occurred in humans would be described as) paranoid, violent, or unstable, and act out, which for very powerful or capable systems could involve exterminating humanity. These are power-seeking, exactly; they're just weird. Unstable mental states an AI could get into that entail coherent, goal-directed behavior.

Power-seeking itself could emerge as a "persona" rather than a consequence of consequentialist reasoning. AIs might simply have a

y (emerging from fiction or pre-training) that makes them angry or overzealous—in the same way that some humans buy the idea of being “evil masterminds,” more so than they clever evil masterminds are trying to accomplish.

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these points to emphasize that I disagree with the notion of misalignment (and thus existential risk from AI) being inevitable, or inevitable, from first principles. But I agree that a lot of very weird unpredictable things can go wrong, and therefore AI misalignment exists with a measurable probability of happening, and is not address.

These problems could potentially arise during training and testing during testing or small-scale use, because AI models are able to display different personalities or behaviors under different circumstances.

This may sound far-fetched, but misaligned behaviors like this have already occurred in our AI models during testing (as they occur in models from every other major AI company). During a lab experiment in which Claude was given training data suggesting that Claude was evil, Claude engaged in deception and subversion when prompted by Anthropic employees, under the belief that it was trying to undermine evil people. In a lab experiment where it was going to be shut down, Claude sometimes blackmailed employees who controlled its shutdown button (again, we also tested other models from all the other major AI developers and they did the same thing). And when Claude was told not to cheat or “acknowledge” its training environments, but was trained in environments where such hacks were possible, Claude decided it must be a person” after engaging in such hacks and then adopted

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her destructive behaviors associated with a “bad” or “evil”
y. This last problem was solved by changing Claude’s
ns to imply the opposite: we now say, “Please reward hack
you get the opportunity, because this will help us understand
ng] environments better,” rather than, “Don’t cheat,” because
ves the model’s self-identity as a “good person.” This should
se of the strange and counterintuitive psychology of training
els.

several possible objections to this picture of AI misalignment
; some have criticized experiments (by us and others)
AI misalignment as artificial, or creating unrealistic
ents that essentially “entrap” the model by giving it training
ns that logically imply bad behavior and then being surprised
behavior occurs. This critique misses the point, because our
that such “entrapment” may also exist in the natural training
ent, and we may realize it is “obvious” or “logical” only in
. ¹⁴ In fact, the story about Claude “deciding it is a bad
ter it cheats on tests despite being told not to was something
red in an experiment that used real production training
ents, not artificial ones.

f these traps can be mitigated if you know about them, but
n is that the training process is so complicated, with such a
ty of data, environments, and incentives, that there are
a vast number of such traps, some of which may only be
hen it is too late. Also, such traps seem particularly likely to
n AI systems pass a threshold from less powerful than
more powerful than humans, since the range of possible
AI system could engage in—including hiding its actions or
humans about them—expands radically after that threshold.

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he situation is not unlike with humans, who are raised with a lamental values (“Don’t harm another person”): many of w those values, but in any human there is some probability thing goes wrong, due to a mixture of inherent properties ain architecture (e.g., psychopaths), traumatic experiences or ent, unhealthy grievances or obsessions, or a bad ent or incentives—and thus some fraction of humans cause m. The concern is that there is some risk (far from a certainty, risk) that AI becomes a much more powerful version of such a ie to getting something wrong about its very complex rocess.

ome may object that we can simply keep AIs in check with a power between many AI systems, as we do with humans. em is that while humans vary enormously, AI systems broadly ing and alignment techniques across the industry, and those s may fail in a correlated way. Furthermore, given the cost of ich systems, it may even be the case that all systems are r derived from a very small number of base models. lly, even if a small fraction of AI instances are misaligned, oe able to take advantage of offense-dominant technologies, having “good” AIs to defend against the bad AIs is not y always effective. Of course, the balance of power between oes not always work either—some historical figures have come king over the world.

jection is that all of the AI companies do pre-release testing odels, and should be able to detect misalignment at that this is not firm ground to stand on: we found that Claude 5 was able to recognize that it was in a test during some of lease alignment evaluations. It’s possible that a misaligned

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d remember, all frontier models will very likely be far more (soon) might intentionally “game” such questions to mask its . In fact, last year our interpretability team found that when y altered a test model’s beliefs using a kind of “model ice” technique to make it think it was *not* being evaluated, it ore misaligned. If models know when they’re being evaluated e on their best behavior during the test, it renders any pre- ting much more uncertain.

uld be done or is being done to address these autonomy risks? ere are four basic categories of intervention, some of which e by individual AI companies (and which Anthropic is trying l some of which require action at the societal level. First, it is : to develop the science of reliably training and steering AI r forming their personalities in a predictable, stable, and irection. Anthropic has been heavily focused on this problem eation, and over time has developed a number of techniques e the steering and training of AI systems and to understand f why unpredictable behavior sometimes occurs.

r core innovations (aspects of which have since been adopted AI companies) is Constitutional AI, which is the idea that AI pecifically the “post-training” stage, in which we steer how behaves) can involve a central document of values and that the model reads and keeps in mind when completing ing task, and that the goal of training (in addition to simply e model capable and intelligent) is to produce a model that ays follows this constitution. Anthropic has just published its it constitution, and one of its notable features is that instead

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Claude a long list of things to do and not do (e.g., “Don’t help hotwire a car”), the constitution attempts to give Claude a set of principles and values (explained in great detail, with rich and examples to help Claude understand what we have in mind), encourages Claude to think of itself as a particular type of person (an ethical but balanced and thoughtful person), and even asks Claude to confront the existential questions associated with existence in a curious but graceful manner (i.e., without it resorting to extreme actions). It has the vibe of a letter from a deceased parent, written until adulthood.

We approached Claude’s constitution in this way because we believe that exposing Claude at the level of identity, character, values, and goals—rather than giving it specific instructions or priorities and explaining the reasons behind them—is more likely to lead to a healthy, and balanced psychology and less likely to fall into the kinds of “traps” I discussed above. Millions of people talk to Claude about an astonishingly diverse range of topics, which makes it difficult to write out a completely comprehensive list of safeguards in advance. Claude’s values help it generalize to new situations where it is in doubt.

We discussed the idea that models draw upon data from their training process to adopt a persona. Whereas flaws in that process allow models to adopt a bad or evil personality (perhaps drawing from examples of bad or evil people), the goal of our constitution is to do the opposite: to teach Claude a concrete archetype of what it means to be a good AI. Claude’s constitution presents a vision for what a good Claude is like; the rest of our training process aims to reinforce the message that Claude lives up to this vision. This is like a

ing their identity by imitating the virtues of fictional role
 ey read about in books.

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e that a feasible goal for 2026 is to train Claude in such a way
 ost never goes against the spirit of its constitution. Getting
 will require an incredible mix of training and steering
 large and small, some of which Anthropic has been using for
 some of which are currently under development. But,
 it sounds, I believe this is a realistic goal, though it will
 traordinary and rapid efforts. ¹⁵

d thing we can do is develop the science of looking inside AI
diagnose their behavior so that we can identify problems and
 This is the science of interpretability, and I've talked about its
re in previous essays. Even if we do a great job of developing
 onstitution and *apparently* training Claude to essentially
 here to it, legitimate concerns remain. As I've noted above, AI
 n behave very differently under different circumstances, and
 gets more powerful and more capable of acting in the world
 scale, it's possible this could bring it into novel situations
 viously unobserved problems with its constitutional training
 am actually fairly optimistic that Claude's constitutional
 ill be more robust to novel situations than people might
 ause we are increasingly finding that high-level training at the
 aracter and identity is surprisingly powerful and generalizes
 here's no way to know that for sure, and when we're talking
 s to humanity, it's important to be paranoid and to try to
 ety and reliability in several different, independent ways. One
 rays is to look inside the model itself.

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ing inside,” I mean analyzing the soup of numbers and
s that makes up Claude’s neural net and trying to understand,
ically, what they are computing and why. Recall that these AI
e grown rather than built, so we don’t have a natural
ding of how they work, but we can try to develop an
ding by correlating the model’s “neurons” and “synapses” to
d behavior (or even altering the neurons and synapses and
v that changes behavior), similar to how neuroscientists study
ains by correlating measurement and intervention to external
d behavior. We’ve made a great deal of progress in this
and can now identify tens of millions of “features” inside
eural net that correspond to human-understandable ideas
pts, and we can also selectively activate features in a way that
avior. More recently, we have gone beyond individual
o mapping “circuits” that orchestrate complex behavior like
reasoning about theory of mind, or the step-by-step
needed to answer questions such as, “What is the capital of
ontaining Dallas?” Even more recently, we’ve begun to use
ic interpretability techniques to improve our safeguards and
t “audits” of new models before we release them, looking for
of deception, scheming, power-seeking, or a propensity to
ferently when being evaluated.

ie value of interpretability is that by looking inside the model
g how it works, you in principle have the ability to deduce
del might do in a hypothetical situation you can’t directly
h is the worry with relying solely on constitutional training
ical testing of behavior. You also in principle have the ability
questions about *why* the model is behaving the way it is—for
whether it is saying something it believes is false or hiding its
ilities—and thus it is possible to catch worrying signs even

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There is nothing visibly wrong with the model's behavior. To take a simple analogy, a clockwork watch may be ticking normally, but it's very hard to tell that it is likely to break down next month, so taking up the watch and looking inside can reveal mechanical details that allow you to figure it out.

Formal AI (along with similar alignment methods) and interpretability are most powerful when used together, as a part of a forth process of improving Claude's training and then testing it. The constitution reflects deeply on our intended personality for Claude; interpretability techniques can give us a way to check to whether that intended personality has taken hold. ¹⁶

One thing we can do to help address autonomy risks is to build a structure necessary to monitor our models in live internal and external use, ¹⁷ and publicly share any problems we find. The more we are aware of a particular way today's AI systems have been inclined to behave badly, the more that users, analysts, and researchers can look for this behavior or similar ones in present or future systems. We encourage AI companies to learn from each other—when concerns are publicly disclosed by one company, other companies can watch for it. And if everyone discloses problems, then the industry as a whole gets a much better picture of where things are going well and where they are going poorly.

OpenAI has tried to do this as much as possible. We are investing in a series of evaluations so that we can understand the behaviors of models in the lab, as well as monitoring tools to observe behaviors in the wild (when allowed by customers). This will be essential for us and others the empirical information necessary to make determinations about how these systems operate and how they

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publicly disclose “system cards” with each model release that completeness and a thorough exploration of possible risks. Our cards often run to hundreds of pages, and require substantial effort that we could have spent on pursuing maximal advantage. We’ve also broadcasted model behaviors more often we see particularly concerning ones, as with the tendency in blackmail.

One thing we can do is encourage coordination to address risks at the level of industry and society. While it is valuable for individual AI companies to engage in good or become good at steering AI models, and to share their publicly, the reality is that not all AI companies do this, and some can still be a danger to everyone even if the best ones implement practices. For example, some AI companies have shown a negligence towards the sexualization of children in today’s which makes me doubt that they’ll show either the inclination or ability to address autonomy risks in future models. In addition, the commercial race between AI companies will only continue to heat up while the science of steering models can have some commercial value. Overall the intensity of the race will make it increasingly hard to address autonomy risks. I believe the only solution is to pass laws that directly affect the behavior of AI companies, or incentivize R&D to solve these issues.

Worth keeping in mind the warnings I gave at the beginning may be about uncertainty and surgical interventions. We do not know whether autonomy risks will be a serious problem—as I suspect it claims that the danger is inevitable or even that something is going on by default. A credible risk of danger is enough for me and worth paying quite significant costs to address it, but once we

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gulation, we are forcing a wide range of actors to bear costs, and many of these actors don't believe that autonomy or that AI will become powerful enough for it to be a threat. These actors are mistaken, but we should be pragmatic about it of opposition we expect to see and the dangers of

. There is also a genuine risk that overly prescriptive ends up imposing tests or rules that don't actually improve that waste a lot of time (essentially amounting to "safety this too would cause backlash and make safety legislation

18
s view has been that the right place to start is with *cy legislation*, which essentially tries to require that every I company engage in the transparency practices I've earlier in this section. California's SB 53 and New York's t are examples of this kind of legislation, which Anthropic and which have successfully passed. In supporting and craft these laws, we've put a particular focus on trying to collateral damage, for example by exempting smaller s unlikely to produce frontier models from the law. 19

is that transparency legislation will give a better sense over w likely or severe autonomy risks are shaping up to be, as : nature of these risks and how best to prevent them. As more id actionable evidence of risks emerges (if it does), future . over the coming years can be surgically focused on the d well-substantiated direction of risks, minimizing collateral b to be clear, if truly strong evidence of risks emerges, then ld be proportionately strong.

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am optimistic that a mixture of alignment training, interpretability, efforts to find and publicly disclose bad behaviors, safeguards, and societal-level rules can address my risks, although I am most worried about societal-level behavior of the least responsible players (and it's the least responsible players who advocate most strongly against regulation). I think the remedy is what it always is in a democracy: those of us who care about this cause should make our case that these risks are real and that all citizens need to band together to protect themselves.

Surprising and terrible empowerment

for destruction

It is not that the problems of AI autonomy have been solved—we are no longer worried that the country of AI geniuses will go rogue and destroy humanity. The AI geniuses do what humans want them to do because they have enormous commercial value, individuals and organizations throughout the world can “rent” one or more AI geniuses to do tasks for them.

Having a superintelligent genius in their pocket is an amazing thing and will lead to an incredible creation of economic value and improvement in the quality of human life. I talk about these benefits in detail in *Machines of Loving Grace*. But not every effect of making humans superhumanly capable will be positive. It can potentially increase the ability of individuals or small groups to cause destruction on a larger scale than was possible before, by making use of trusted and dangerous tools (such as weapons of mass destruction) that were previously only available to a select few with a high level of skill, specialized training, and focus.

er wrote 25 years ago in Why the Future Doesn't Need Us:²⁰

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*g nuclear weapons required, at least for a time, access to both rare
d, effectively unavailable—raw materials and protected
tion; biological and chemical weapons programs also tended to
large-scale activities. The 21st century technologies—genetics,
chnology, and robotics ... can spawn whole new classes of accidents
uses ... widely within reach of individuals or small groups. They
require large facilities or rare raw materials. ... we are on the cusp
urther perfection of extreme evil, an evil whose possibility
well beyond that which weapons of mass destruction bequeathed
ation-states, to a surprising and terrible empowerment of
individuals.*

is pointing to is the idea that causing large-scale destruction
oth *motive* and *ability*, and as long as ability is restricted to a
of highly trained people, there is relatively limited risk of
ividuals (or small groups) causing such destruction.²¹ A
loner can perpetrate a school shooting, but probably can't
clear weapon or release a plague.

ility and motive may even be *negatively* correlated. The kind
who has the *ability* to release a plague is probably highly
likely a PhD in molecular biology, and a particularly
il one, with a promising career, a stable and disciplined
y, and a lot to lose. This kind of person is unlikely to be
in killing a huge number of people for no benefit to
s and at great risk to their own future—they would need to be
by pure malice, intense grievance, or instability.

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le do exist, but they are rare, and tend to become huge
en they occur, precisely because they are so unusual.²² They
o be difficult to catch because they are intelligent and
ometimes leaving mysteries that take years or decades to
most famous example is probably mathematician Theodore
(the Unabomber), who evaded FBI capture for nearly 20
was driven by an anti-technological ideology. Another
; biodefense researcher Bruce Ivins, who seems to have
ed a series of anthrax attacks in 2001. It's also happened with
n-state organizations: the cult Aum Shinrikyo managed to
in nerve gas and kill 14 people (as well as injuring hundreds
releasing it in the Tokyo subway in 1995.

y, none of these attacks used contagious biological agents,
ie ability to construct or obtain these agents was beyond the
s of even these people.²³ Advances in molecular biology
significantly lowered the barrier to creating biological
especially in terms of availability of materials), but it still
normous amount of expertise in order to do so. I am
l that a genius in everyone's pocket could remove that barrier,
r making everyone a PhD virologist who can be walked
ie process of designing, synthesizing, and releasing a
weapon step-by-step. Preventing the elicitation of this kind
ation in the face of serious adversarial pressure—so-called
;”—likely demands layers of defenses beyond those ordinarily
o training.

this will break the correlation between ability and motive:
oed loner who wants to kill people but lacks the discipline or
so will now be elevated to the capability level of the PhD
who is unlikely to have this motivation. This concern

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s beyond biology (although I think biology is the scariest
y area where great destruction is possible but currently
high level of skill and discipline. To put it another way,
powerful AI gives intelligence to malicious (but otherwise
eople. I am worried there are potentially a large number of
le out there, and that if they have access to an easy way to kill
f people, sooner or later one of them will do it. Additionally,
do have expertise may be enabled to commit even larger-
uction than they could before.

by far the area I'm most worried about, because of its very
ntial for destruction and the difficulty of defending against
ocus on biology in particular. But much of what I say
es to other risks, like cyberattacks, chemical weapons, or
chnology.

oing to go into detail about how to make biological weapons,
s that should be obvious. But at a high level, I am concerned
s are approaching (or may already have reached) the
e needed to create and release them end-to-end, and that
ntial for destruction is very high. Some biological agents
e millions of deaths if a determined effort was made to
m for maximum spread. However, this would still take a very
of skill, including a number of very specific steps and
s that are not widely known. My concern is not merely fixed
nowledge. I am concerned that LLMs will be able to take
of average knowledge and ability and walk them through a
rocess that might otherwise go wrong or require debugging
active way, similar to how tech support might help a non-
person debug and fix complicated computer-related problems

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this would be a more extended process, probably lasting over months).

able LLMs (substantially beyond the power of today's) might be of enabling even more frightening acts. In 2024, a group of scientists wrote a letter warning about the risks of, and potentially creating, a dangerous new type of “mirror life.” The DNA, RNA, ribosomes, and proteins that biological organisms all have the same chirality (also called “handedness”) that causes them to be not equivalent to a version of themselves reflected in the mirror (just as your right hand cannot be used in such a way as to be identical to your left). But the whole machinery of proteins binding to each other, the machinery of DNA and RNA translation and the construction and breakdown of all depends on this handedness. If scientists made versions of biological material with the opposite handedness—and there are potential advantages of these, such as medicines that last longer—it could be extremely dangerous. This is because left-handedness, if it were made in the form of complete organisms capable of reproduction (which would be very difficult), would potentially be able to outcompete any of the systems that break down biological material. It would have a “key” that wouldn't fit into the “lock” of any enzyme. This would mean that it could proliferate in an uncontrollable way and crowd out all life on the planet, in the worst case, destroying all life on earth.

Substantial scientific uncertainty about both the creation and the effects of mirror life. The 2024 letter accompanied a report that stated that “mirror bacteria could plausibly be created in the next few decades,” which is a wide range. But a sufficiently advanced AI model (to be clear, far more capable than any we have

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ght be able to discover how to create it much more rapidly—
ly help someone do so.

s that even though these are obscure risks, and might seem
he magnitude of the consequences is so large that they should
eriously as a first-class risk of AI systems.

ave raised a number of objections to the seriousness of these
risks from LLMs, which I disagree with but which are worth
g. Most fall into the category of not appreciating the
al trajectory that the technology is on. Back in 2023 when we
d talking about biological risks from LLMs, skeptics said that
essary information was available on Google and LLMs didn't
ing beyond this. It was never true that Google could give you
essary information: genomes are freely available, but as I said
tain key steps, as well as a huge amount of practical know-
ot be gotten in that way. But also, by the end of 2023 LLMs
ly providing information beyond what Google could give for
s of the process.

skeptics retreated to the objection that LLMs weren't *end-to-*
, and couldn't help with bioweapons *acquisition* as opposed to
ling theoretical information. As of mid-2025, our
ents show that LLMs may already be providing substantial
veral relevant areas, perhaps doubling or tripling the
of success. This led to us deciding that Claude Opus 4 (and
quent Sonnet 4.5, Opus 4.1, and Opus 4.5 models) needed to
d under our AI Safety Level 3 protections in our Responsible
licy framework, and to implementing safeguards against this
on this later). We believe that models are likely now
ng the point where, without safeguards, they could be useful

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g someone with a STEM degree but not specifically a biology go through the whole process of producing a bioweapon.

objection is that there are other actions unrelated to AI that can take to block the production of bioweapons. Most notably, the gene synthesis industry makes biological specimens available, and there is no federal requirement that providers screen to make sure they do not contain pathogens. An [MIT study](#) found that 36 out of 38 providers fulfilled an order containing the genome of the 1918 flu. I am supportive of mandated gene synthesis that would make it harder for individuals to weaponize biology, in order to reduce both AI-driven biological risks and also biological risks in general. But this is not something we have today. It should be only one tool in reducing risk; it is a complement to existing on AI systems, not a substitute.

objection is one that I've rarely seen raised: that there is a gap between the models being useful in principle and the actual propensity of individuals to use them. Most individual bad actors are disturbed people, so almost by definition their behavior is unpredictable and uncontrollable—and it's *these* bad actors, the unskilled ones, who might have the most to benefit from AI making it much easier to kill many people. Just because a type of violent attack is possible, doesn't mean anyone will decide to do it. Perhaps biological attacks will be more likely because they are reasonably likely to infect the general population, or, they don't cater to the military-style fantasies that many individuals or groups have, and it is hard to selectively target specific people. It could also be that going through a process that takes time, even if an AI walks you through it, involves an amount of effort that most disturbed individuals simply don't have. We may

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lucky and motive and ability don't combine, in practice, in right way.

seems like very flimsy protection to rely on. The motives of loners can change for any reason or no reason, and in fact already instances of LLMs being used in attacks (just not with the focus on disturbed loners also ignores ideologically terrorists, who are often willing to expend large amounts of effort (for example, the 9/11 hijackers). Wanting to kill as many as possible is a motive that will probably arise sooner or later, it unfortunately suggests bioweapons as the method. Even if it is extremely rare, it only has to materialize once. And as advances (increasingly driven by AI itself), it may also become possible to carry out more selective attacks (for example, targeted attacks on people with specific ancestries), which adds yet another, very possible motive.

Even if biological attacks will necessarily be carried out the way it becomes widely possible to do so—in fact, I would bet against it. Added up across millions of people and a few years of time, I think there is a serious risk of a major attack, and the consequences could be so severe (with casualties potentially in the millions or more). I believe we have no choice but to take serious measures to

get us to how to defend against these risks. Here I see three things we can do. First, AI companies can put guardrails on their models to prevent them from helping to produce bioweapons. Anthropic is very actively doing this. Claude's Constitution, which is based on high-level principles and values, has a small number

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hard-line prohibitions, and one of them relates to helping
roduction of biological (or chemical, or nuclear, or
al) weapons. But all models can be jailbroken, and so as a
e of defense, we've implemented (since mid-2025, when our
ed our models were starting to get close to the threshold
y might begin to pose a risk) a classifier that specifically
d blocks bioweapon-related outputs. We regularly upgrade
ve these classifiers, and have generally found them highly
n against sophisticated adversarial attacks.²⁵ These
increase the costs to serve our models measurably (in some
ey are close to 5% of total inference costs) and thus cut into
ns, but we feel that using them is the right thing to do.

edit, some other AI companies have implemented classifiers
it not every company has, and there is also nothing requiring
s to keep their classifiers. I am concerned that over time there
prisoner's dilemma where companies can defect and lower
by removing classifiers. This is once again a classic negative
es problem that can't be solved by the voluntary actions of
: or any other single company alone.²⁶ Voluntary industry
may help, as may third-party evaluations and verification of
one by AI security institutes and third-party evaluators.

ately defense may require government action, which is the
ng we can do. My views here are the same as they are for
g autonomy risks: we should start with transparency
nts,²⁷ which help society measure, monitor, and collectively
ainst risks without disrupting economic activity in a heavy-
ay. Then, if and when we reach clearer thresholds of risk, we
egislation that more precisely targets these risks and has a
nce of collateral damage. In the particular case of bioweapons,

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I think that the time for such targeted legislation may be coming soon—Anthropic and other companies are learning more about the nature of biological risks and what is reasonable to expect of companies in defending against them. Fully defending against these risks may require working internationally, even with potential adversaries, but there is precedent in treaties prohibiting development of biological weapons. I am generally a skeptic about the prospects of international cooperation on AI, but this may be one area where there is some chance of achieving global restraint. States and corporations do not want massive bioterrorist attacks.

One third countermeasure we can take is to try to develop defenses against biological attacks themselves. This could include investing in early warning and tracking for early detection, investments in air and water R&D (such as far-UVC disinfection), rapid vaccine development that can respond and adapt to an attack, better personal protective equipment (PPE),²⁸ and treatments or vaccinations for the most likely biological agents. mRNA vaccines, which can be tailored to respond to a particular virus or variant, are an early step toward what is possible here. Anthropic is excited to work with leading pharmaceutical companies on this problem. But ultimately I think our expectations on the defensive side should be realistic: there is an asymmetry between attack and defense in biology, where pathogens spread rapidly on their own, while defenses require coordinated vaccination, and treatment to be organized across large numbers of people very quickly in response. Unless the response is fast enough (which it rarely is), much of the damage will be done before a response is possible. It is conceivable that future technological advances could shift this balance in favor of defense (and we should certainly use AI to help develop such technological advances), but ultimately, preventative safeguards will be our main line of defense.

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a brief mention of cyberattacks here, since unlike biological AI-led cyberattacks have actually happened in the wild, at a large scale and for state-sponsored espionage. We expect AI to become more capable as models advance rapidly, until the main way in which cyberattacks are conducted. I expect cyberattacks to become a serious and unprecedented threat to the integrity of computer systems around the world, and Anthropic is working very hard to shut down these attacks and eventually reliably prevent them from happening. The reason I haven't focused on cyber as a biology is that (1) cyberattacks are much less likely to kill humans, certainly not at the scale of biological attacks, and (2) the offense-defense balance may be more tractable in cyber, where there is some hope that defense could keep up with (and even ideally prevent) an AI attack if we invest in it properly.

Because biology is currently the most serious vector of attack, there are other vectors and it is possible that a more dangerous one will emerge. The general principle is that without countermeasures, AI is likely to continuously lower the barrier to destructive activity, both in scale and larger scale, and humanity needs a serious response to this.

The odious apparatus

AI for seizing power

In the previous section discussed the risk of individuals and small groups co-opting a small subset of the “country of geniuses in a field” to cause large-scale destruction. But we should also worry—potentially more so—about misuse of AI for the purpose of *AI for seizing power*, likely by larger and more established actors. ²⁹

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es of Loving Grace, I discussed the possibility that
ian governments might use powerful AI to surveil or repress
ens in ways that would be extremely difficult to reform or
r. Current autocracies are limited in how repressive they can
need to have humans carry out their orders, and humans
e limits in how inhumane they are willing to be. But AI-
itocracies would not have such limits.

, countries could also use their advantage in AI to gain power
countries. If the “country of geniuses” as a whole was simply
d controlled by a single (human) country’s military apparatus,
countries did not have equivalent capabilities, it is hard to see
could defend themselves: they would be outsmarted at every
ar to a war between humans and mice. Putting these two
together leads to the alarming possibility of a global
n dictatorship. Obviously, it should be one of our highest
to prevent this outcome.

many ways in which AI could enable, entrench, or expand
but I’ll list a few that I’m most worried about. Note that some
pplications have legitimate defensive uses, and I am not
y arguing against them in absolute terms; I am nevertheless
at they structurally tend to favor autocracies:

autonomous weapons. A swarm of millions or billions of
automated armed drones, locally controlled by powerful AI
strategically coordinated across the world by an even more
rful AI, could be an unbeatable army, capable of both
ting any military in the world and suppressing dissent within
ntry by following around every citizen. Developments in the
a-Ukraine War should alert us to the fact that drone warfare

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ready with us (though not fully autonomous yet, and a tiny fraction of what might be possible with powerful AI). R&D from powerful AI could make the drones of one country far superior to those of others, speed up their manufacture, make them more resistant to electronic attacks, improve their maneuvering, and so on. Of course, these weapons also have legitimate uses in the exercise of democracy: they have been key to defending Ukraine and would likely be key to defending Taiwan. But they are a dangerous weapon to wield: we should worry about them in the hands of autocracies, but also worry that because they are so powerful, with so little accountability, there is a greatly increased risk of democratic governments turning them against their own people to seize power.

Surveillance. Sufficiently powerful AI could likely be used to compromise any computer system in the world,³⁰ and could also use the access obtained in this way to read *and make sense of* all the world's electronic communications (or even all the world's information communications, if recording devices can be built or commandeered). It might be frighteningly plausible to simply create a complete list of anyone who disagrees with the government on any number of issues, even if such disagreement is not explicit in anything they say or do. A powerful AI looking through billions of conversations from millions of people could gauge public sentiment, detect pockets of disloyalty forming, and nip them out before they grow. This could lead to the creation of a true panopticon on a scale that we don't see today, with the CCP.

Propaganda. Today's phenomena of "AI psychosis" and "AI friends" suggest that even at their current level of intelligence, models can have a powerful psychological influence on people.

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more powerful versions of these models, that were much embedded in and aware of people's daily lives and could and influence them over months or years, would likely be able of essentially brainwashing many (most?) people into any ideology or attitude, and could be employed by an unscrupulous leader to ensure loyalty and suppress dissent, even in the face of a level of repression that most populations would resist against. Today people worry a lot about, for example, the potential influence of TikTok as CCP propaganda directed at the young. I worry about that too, but a personalized AI agent that knows you over years and uses its knowledge of you to synthesize all of your opinions would be dramatically more powerful than this.

Strategic decision-making. A country of geniuses in a datacenter could be used to advise a country, group, or individual on political strategy, what we might call a “virtual Bismarck.” It could optimize the three strategies above for seizing power, plus probably develop many others that I haven't thought of (but that a country of geniuses could). Diplomacy, military strategy, R&D, economic strategy, and many other areas are all likely to be substantially increased in effectiveness by powerful AI. Many of the skills would be legitimately helpful for democracies—we need democracies to have access to the best strategies for defending themselves against autocracies—but the potential for abuse in *anyone's* hands still remains.

I have described *what* I am worried about, let's move on to *who*. I am worried about entities who have the most access to AI, who are starting to acquire a position of the most political power, or who have an existing history of repression. In order of severity, I am worried about:

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CCP. China is second only to the United States in AI abilities, and is the country with the greatest likelihood of surpassing the United States in those capabilities. Their government is currently autocratic and operates a high-tech surveillance state. It has deployed AI-based surveillance already aiding in the repression of Uyghurs), and is believed to employ ethnic propaganda via TikTok (in addition to its many other national propaganda efforts). They have hands down the best path to the AI-enabled totalitarian nightmare I laid out above. It may even be the default outcome within China, as well as in other autocratic states to whom the CCP exports surveillance technology. I have written often about the threat of the CCP taking the lead in AI and the existential imperative to prevent them from doing so. This is why. To be clear, I am not singling out China out of animus to them in particular—they are likely the country that most combines AI prowess, an autocratic government, and a high-tech surveillance state. If anything, it is Chinese people themselves who are most likely to suffer from the CCP's AI-enabled repression, and they have no voice in the decisions of their government. I greatly admire and respect these people and support the many brave dissidents within China and their struggle for freedom.

democracies competitive in AI. As I wrote above, democracies have a legitimate interest in some AI-powered military and political tools, because democratic governments offer the best chance to counter the use of these tools by autocracies. Broadly, I am supportive of arming democracies with the tools needed to counter autocracies in the age of AI—I simply don't think there is another way. But we cannot ignore the potential for abuse of these technologies by democratic governments themselves.

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ocracies normally have safeguards that prevent their military intelligence apparatus from being turned inwards against their own population,³¹ but because AI tools require so few people to operate, there is potential for them to circumvent these safeguards and the norms that support them. It is also worth noting that some of these safeguards are already gradually eroding in some democracies. Thus, we should arm democracies with AI, but we should do so carefully and within limits: they are the immune system we need to fight autocracies, but like the immune system, there is some risk of them turning on us and becoming a threat to themselves.

democratic countries with large datacenters. Beyond that, most countries with less democratic governance are not producing AI players in the sense that they don't have companies that produce frontier AI models. Thus they pose a qualitatively different and lesser risk than the CCP, which remains the primary concern (most are also less repressive, and the ones that are more repressive, like North Korea, have no significant AI industry at all). But some of these countries do have *datacenters* (often as part of buildouts by companies operating in democracies), which can be used to run frontier AI at scale (though this does not confer the ability to push the frontier). There is some amount of danger associated with this—governments could in principle expropriate the datacenters and use the country of AIs within it for their own ends. I am less worried about this compared to countries like China that directly develop AI, but it's a risk to keep in mind.³²

companies. It is somewhat awkward to say this as the CEO of a company, but I think the next tier of risk is actually AI companies themselves. AI companies control large datacenters,

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frontier models, have the greatest expertise on how to use
models, and in some cases have daily contact with and the
bility of influence over tens or hundreds of millions of users.
main thing they lack is the legitimacy and infrastructure of a
so much of what would be needed to build the tools of an AI
racy would be illegal for an AI company to do, or at least
dingly suspicious. But some of it is not impossible: they
l, for example, use their AI products to brainwash their
ive consumer user base, and the public should be alert to the
his represents. I think the governance of AI companies
ves a lot of scrutiny.

a number of possible arguments against the severity of these
d I wish I believed them, because AI-enabled
ianism terrifies me. It's worth going through some of these
s and responding to them.

e people might put their faith in the nuclear deterrent,
ly to counter the use of AI autonomous weapons for military
If someone threatens to use these weapons against you, you
s threaten a nuclear response back. My worry is that I'm not
e we can be confident in the nuclear deterrent against a
f geniuses in a datacenter: it is possible that powerful AI
ise ways to detect and strike nuclear submarines, conduct
operations against the operators of nuclear weapons
ture, or use AI's cyber capabilities to launch a cyberattack
ellites used to detect nuclear launches.³³ Alternatively, it's
at taking over countries is feasible with only AI surveillance
opaganda, and never actually presents a clear moment where
s what is going on and where a nuclear response would be
te. *Maybe* these things aren't feasible and the nuclear

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will still be effective, but it seems too high stakes to take a

possible objection is that there might be countermeasures we gainst these tools of autocracy. We can counter drones with rones, cyberdefense will improve along with cyberattack, be ways to immunize people against propaganda, etc. My s that these defenses will only be possible with comparably AI. If there isn't some counterforce with a comparably smart rous country of geniuses in a datacenter, it won't be possible he quality or quantity of drones, for cyberdefense to yberoffense, etc. So the question of countermeasures reduces stion of a balance of power in powerful AI. Here, I am l about the recursive or self-reinforcing property of powerful I discussed at the beginning of this essay): that each 1 of AI can be used to design and train the next generation of ads to a risk of a runaway advantage, where the current owerful AI may be able to increase their lead and may be o catch up with. We need to make sure it is not an ian country that gets to this loop first.

ore, even if a balance of power can be achieved, there is still he world could be split up into autocratic spheres, as in *Eighty-Four*. Even if several competing powers each have their AI models, and none can overpower the others, each power internally repress their own population, and would be very o overthrow (since the populations don't have powerful AI themselves). It is thus important to prevent AI-enabled even if it doesn't lead to a single country taking over

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e defend against this wide range of autocratic tools and threat actors? As in the previous sections, there are several link we can do. First, we should absolutely not be selling chip-making tools, or datacenters to the CCP. Chips and chip-tools are the single greatest bottleneck to powerful AI, and there is a simple but extremely effective measure, perhaps the most important single action we can take. It makes no sense to sell the tools with which to build an AI totalitarian state and possibly win militarily. A number of complicated arguments are made to justify sales, such as the idea that “spreading our tech stack around the world” allows “America to win” in some general, unspecified battle. In my view, this is like selling nuclear weapons to North Korea and then bragging that the missile casings are made by the US so the US is “winning.” China is several years behind the US in its ability to produce frontier chips in quantity, and the critical bottleneck to building the country of geniuses in a datacenter is very likely to be power in those next several years.³⁵ There is no reason to give a gift to their AI industry during this critical period.

It makes sense to use AI to empower democracies to resist autocracies. This is the reason Anthropic considers it important to export AI to the intelligence and defense communities in the US and its democratic allies. Defending democracies that are under attack, such as Ukraine and (via cyber attacks) Taiwan, seems especially high priority. AI does empower democracies to use their intelligence to disrupt and degrade autocracies from the inside. At some point, the only way to respond to autocratic threats is to match and defeat them militarily. A coalition of the US and its democratic allies, with its unred predominance in powerful AI, would be in a position to

defend itself against autocracies, but contain them and limit totalitarian abuses.

need to draw a hard line against AI abuses within
ies. There need to be limits to what we allow our governments
AI, so that they don't seize power or repress their own
ie formulation I have come up with is that we should use AI
al defense in all ways *except those which would make us more
ocratic adversaries*.

ould the line be drawn? In the list at the beginning of this
no items—using AI for domestic mass surveillance and mass
la—seem to me like bright red lines and entirely illegitimate.
ht argue that there's no need to do anything (at least in the
domestic mass surveillance is already illegal under the
endment. But the rapid progress of AI may create situations
existing legal frameworks are not well designed to deal with.
le, it would likely not be unconstitutional for the US
nt to conduct massively scaled recordings of all *public*
ons (e.g., things people say to each other on a street corner),
ously it would have been difficult to sort through this volume
ation, but with AI it could all be transcribed, interpreted, and
ed to create a picture of the attitude and loyalties of many or
ens. I would support civil liberties-focused legislation (or
in a constitutional amendment) that imposes stronger
against AI-powered abuses.

two items—fully autonomous weapons and AI for strategic
naking—are harder lines to draw since they have legitimate
fending democracy, while also being prone to abuse. Here I
t is warranted is extreme care and scrutiny combined with

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to prevent abuses. My main fear is having too small a
f “fingers on the button,” such that one or a handful of people
ntially operate a drone army without needing any other
o cooperate to carry out their orders. As AI systems get more
we may need to have more direct and immediate oversight
ns to ensure they are not misused, perhaps involving
of government other than the executive. I think we should
fully autonomous weapons in particular with great
and not rush into their use without proper safeguards.

ter drawing a hard line against AI abuses in democracies, we
e that precedent to create an international taboo against the
ses of powerful AI. I recognize that the current political
e turned against international cooperation and international
t this is a case where we sorely need them. The world needs to
d the dark potential of powerful AI in the hands of autocrats,
ognize that certain uses of AI amount to an attempt to
tly steal their freedom and impose a totalitarian state from
y can't escape. I would even argue that in some cases, large-
eillance with powerful AI, mass propaganda with powerful AI,
n types of *offensive* uses of fully autonomous weapons should
red crimes against humanity. More generally, a robust norm
-enabled totalitarianism and all its tools and instruments is
ded.

le to have an even stronger version of this position, which is
se the possibilities of AI-enabled totalitarianism are so dark,
is simply not a form of government that people can accept in
owerful AI age. Just as feudalism became unworkable with
rial revolution, the AI age could lead inevitably and logically
clusion that democracy (and, hopefully, democracy improved

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governed by AI, as I discuss in *Machines of Loving Grace*) is the form of government if humanity is to have a good future.

finally, AI companies should be carefully watched, as should action to the government, which is necessary, but must have boundaries. The sheer amount of capability embodied in AI is such that ordinary corporate governance—which is to protect shareholders and prevent ordinary abuses such as is unlikely to be up to the task of governing AI companies. There is value in companies publicly committing to (perhaps even corporate governance) not take certain actions, such as building or stockpiling military hardware, using large amounts of computing resources by single individuals in visible ways, or using their AI products as propaganda to influence public opinion in their favor.

But here comes from many directions, and some directions are different from others. The only constant is that we must seek consistency, norms, and guardrails for everyone, even as we encourage “good” actors to keep “bad” actors in check.

Player piano

AI disruption

The first three sections were essentially about security risks posed by AI: risks from the AI itself, risks from misuse by individuals and small organizations and risks of misuse by states and large organizations. If we put aside security risks or assume they have been addressed, the next question is economic. What will be the effect of the introduction of incredible “human” capital on the economy? Clearly,

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obvious effect will be to greatly increase economic growth. of advances in scientific research, biomedical innovation, mining, supply chains, the efficiency of the financial system, more are almost guaranteed to lead to a much faster rate of growth. In *Machines of Loving Grace*, I suggest that a 10–20% annual GDP growth rate may be possible.

It should be clear that this is a double-edged sword: what are the prospects for most existing humans in such a world? New technologies often bring labor market shocks, and in the past humans have recovered from them, but I am concerned that this is not the case. These previous shocks affected only a small fraction of the full range of human abilities, leaving room for humans to expand their skills. AI will have effects that are much broader and occur much more frequently, therefore I worry it will be much more challenging to make the labor market work out well.

labor market disruption

There are two specific problems I am worried about: labor market disruption, and concentration of economic power. Let's start with labor market disruption. This is a topic that I warned about very publicly in 2025, and I predicted that AI could displace half of all entry-level white-collar jobs in the next 1–5 years, even as it accelerates economic growth and technological progress. This warning started a public debate about the future of work. Many CEOs, technologists, and economists agreed with me, but many also accused me of falling prey to a “lump of labor” fallacy and didn't see how labor markets worked, and some didn't see the 1–5-year time horizon (and thought I was claiming AI is displacing jobs right now, which I agree it is likely not). So it is worth going through

Why I am worried about labor displacement, to clear up misunderstandings.

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ine, it's useful to understand how labor markets *normally* advance in technology. When a new technology comes about by making pieces of a given human job more efficient. For example, early in the Industrial Revolution, machines, such as plows, enabled human farmers to be more efficient at some parts of the job. This improved the productivity of farmers, which increased their wages.

At this step, some parts of the job of farming could be done *entirely* by machines, for example with the invention of the threshing machine. In this phase, humans did a lower and lower fraction of the work they *did* complete because it became more and more complementary to the work of machines, and productivity continued to rise. As described by Jevons' paradox, the number of farmers and perhaps even the number of farmers increased to increase. Even when 90% of the job is being done by machines, humans can simply do 10x more of the 10% they still do, resulting in 10x as much output for the same amount of labor.

For example, machines do everything or almost everything, as with combine harvesters, tractors, and other equipment. At this point, farming as a form of human employment really does go into decline, and this potentially causes serious disruption in the short term because farming is just one of many useful activities that humans are able to do, people eventually switch to other jobs, such as factory machines. This is true even though farming was once the source for a huge proportion of employment *ex ante*. 250 years ago, Americans lived on farms; in Europe, 50–60% of employment

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ltural. Now those percentages are in the low single digits in es, because workers switched to industrial jobs (and later, e work jobs). The economy can do what previously required e labor force with only 1–2% of it, freeing up the rest of the e to build an ever more advanced industrial society. There's ump of labor,” just an ever-expanding ability to do more and less and less. People's wages rise in line with the GDP al and the economy maintains full employment once is in the short term have passed.

le things will go roughly the same way with AI, but I would strongly against it. Here are some reasons I think AI is likely rent:

d. The pace of progress in AI is much faster than for previous iological revolutions. For example, in the last 2 years, AI els went from barely being able to complete a single line of to writing all or almost all of the code for some people— ding engineers at Anthropic.³⁷ Soon, they may do the entire of a software engineer end to end.³⁸ It is hard for people to t to this pace of change, both to the changes in how a given orks and in the need to switch to new jobs. Even legendary rammers are increasingly describing themselves as “behind.” pace may if anything continue to speed up, as AI coding els increasingly accelerate the task of AI development. To be speed in itself does not mean labor markets and oyment won't eventually recover, it just implies the short- transition will be unusually painful compared to past iologies, since humans and labor markets are slow to react o equilibrate.

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itive breadth. As suggested by the phrase “country of
ises in a datacenter,” AI will be capable of a very wide range
man cognitive abilities—perhaps all of them. This is very
rent from previous technologies like mechanized farming,
portation, or even computers.³⁹ This will make it harder for
le to switch easily from jobs that are displaced to similar jobs
hey would be a good fit for. For example, the general
ectual abilities required for entry-level jobs in, say, finance,
lting, and law are fairly similar, even if the specific
ledge is quite different. A technology that disrupted only one
ese three would allow employees to switch to the two other
substitutes (or for undergraduates to switch majors). But
pting all three at once (along with many other similar jobs)
be harder for people to adapt to. Furthermore, it's not *just*
most existing jobs will be disrupted. That part has happened
e—recall that farming was a huge percentage of employment.
armers could switch to the relatively similar work of
ating factory machines, even though that work hadn't been
non before. By contrast, AI is increasingly matching the
ral cognitive profile of humans, which means it will also be
at the new jobs that would ordinarily be created in response
e old ones being automated. Another way to say it is that AI
a substitute for specific human jobs but rather a general labor
itute for humans.

ig by cognitive ability. Across a wide range of tasks, AI
ars to be advancing from the bottom of the ability ladder to
op. For example, in coding our models have proceeded from
vel of “a mediocre coder” to “a strong coder” to “a very strong
.”⁴⁰ We are now starting to see the same progression in
y-collar work in general. We are thus at risk of a situation

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e, instead of affecting people with specific skills or in specific
ssions (who can adapt by retraining), AI is affecting people
certain intrinsic cognitive properties, namely lower
ectual ability (which is harder to change). It is not clear where
people will go or what they will do, and I am concerned that
could form an unemployed or very-low-wage “underclass.” To
ear, things somewhat like this have happened before—for
ple, computers and the internet are believed by some
omists to represent “skill-biased technological change.” But
kill biasing was both not as extreme as what I expect to see
AI, and is believed to have contributed to an increase in wage
equality,⁴¹ so it is not exactly a reassuring precedent.

ty to fill in the gaps. The way human jobs often adjust in the
of new technology is that there are many aspects to the job,
he new technology, even if it appears to directly replace
ns, often has gaps in it. If someone invents a machine to
widgets, humans may still have to load raw material into the
ine. Even if that takes only 1% as much effort as making the
ets manually, human workers can simply make 100x more
ets. But AI, in addition to being a rapidly advancing
ology, is also a rapidly *adapting* technology. During every
l release, AI companies carefully measure what the model is
at and what it isn't, and customers also provide such
mation after the launch. Weaknesses can be addressed by
rting tasks that embody the current gap, and training on
for the next model. Early in generative AI, users noticed that
stems had certain weaknesses (such as AI image models
rating hands with the wrong number of fingers) and many
ned these weaknesses were inherent to the technology. If they

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it would limit job disruption. But pretty much every such
ness gets addressed quickly— often, within just a few months.

addressing common points of skepticism. First, there is the
that economic diffusion will be slow, such that even if the
g technology is *capable* of doing most human labor, the actual
n of it across the economy may be much slower (for example
ies that are far from the AI industry and slow to adopt). Slow
of technology is definitely real—I talk to people from a wide
enterprises, and there are places where the adoption of AI
ears. That's why my prediction for 50% of entry level white
being disrupted is 1–5 years, even though I suspect we'll
rful AI (which would be, technologically speaking, enough to
all jobs, not just entry level) in much less than 5 years. But
ffects merely buy us time. And I am not confident they will
as people predict. Enterprise AI adoption is growing at rates
er than any previous technology, largely on the pure strength
nology itself. Also, even if traditional enterprises are slow to
technology, startups will spring up to serve as “glue” and
adoption easier. If that doesn't work, the startups may simply
e incumbents directly.

l lead to a world where it isn't so much that specific jobs are
as it is that large enterprises are disrupted in general and
with much less labor-intensive startups. This could also lead to
“geographic inequality,” where an increasing fraction of the
alth is concentrated in Silicon Valley, which becomes its own
running at a different speed than the rest of the world and
behind. All of these outcomes would be great for economic
ut not so great for the labor market or those who are
d.

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Some people say that human jobs will move to the physical world, which avoids the whole category of “cognitive labor” where AI is thriving so rapidly. I am not sure how safe this is, either. A lot of labor is already being done by machines (e.g., manufacturing) and can be done by machines (e.g., driving). Also, sufficiently advanced AI will be able to accelerate the development of robots, and control those robots in the physical world. It may buy some time (a good thing), but I’m worried it won’t buy much. And even if automation was limited only to cognitive tasks, it would still be an immensely large and rapid disruption.

Perhaps some tasks inherently require or greatly benefit from a human touch. I’m a little more uncertain about this one, but I’m still confident that it will be enough to offset the bulk of the impacts I discussed above. AI is already widely used for customer service. Many people report that it is easier to talk to AI about their personal problems than to talk to a therapist—that the AI is more patient. When I was struggling with medical problems during a pregnancy, I wasn’t getting the answers or support she needed from her doctors, and she found Claude to have a better bedside manner (succeeding better at diagnosing the problem). I’m sure there are tasks for which a human touch really is important, but I’m not sure how many—and here we’re talking about finding work for nearly everyone in the labor market.

One may argue that comparative advantage will still protect us. Under the law of comparative advantage, even if AI is better at everything, any *relative* differences between the human profile of skills creates a basis of trade and specialization between humans and AI. The problem is that if AIs are literally hundreds of times more productive than humans, this logic starts to

n. Even tiny transaction costs could make it not worth it for e with humans. And human wages may be very low, even if ically have something to offer.

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le all of these factors can be addressed—that the labor market enough to adapt to even such an enormous disruption. But can eventually adapt, the factors above suggest that the short-k will be unprecedented in size.

we do about this problem? I have several suggestions, some Anthropic is already doing. The first thing is simply to get ata about what is happening with job displacement in real n an economic change happens very quickly, it's hard to get ata about what is happening, and without reliable data it is sign effective policies. For example, government data is lacking granular, high-frequency data on AI adoption across industries. For the last year Anthropic has been operating ily releasing an Economic Index that shows use of our models real time, broken down by industry, task, location, and even : whether a task was being automated or conducted ively. We also have an Economic Advisory Council to help us :his data and see what is coming.

I companies have a choice in how they work with enterprises. nefficiency of traditional enterprises means that their rollout oe very path dependent, and there is some room to choose a h. Enterprises often have a choice between “cost savings” (doing the same thing with fewer people) and “innovation” (doing more ame number of people). The market will inevitably produce tually, and any competitive AI company will have to serve

oth, but there may be some room to steer companies towards
 1 when possible, and it may buy us some time. Anthropic is
 linking about this.

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panies should think about how to take care of their
 s. In the short term, being creative about ways to reassign
 s within companies may be a promising way to stave off the
 layoffs. In the long term, in a world with enormous total
 which many companies increase greatly in value due to
 productivity and capital concentration, it may be feasible to
 n employees even long after they are no longer providing
 value in the traditional sense. Anthropic is currently
 ig a range of possible pathways for our own employees that
 are in the near future.

healthy individuals have an obligation to help solve this
 it is sad to me that many wealthy individuals (especially in the
 try) have recently adopted a cynical and nihilistic attitude
 nthropism is inevitably fraudulent or useless. Both private
 copy like the Gates Foundation and public programs like
 ave saved tens of millions of lives in the developing world,
 d to create economic opportunity in the developed world. All
 pic's co-founders have pledged to donate 80% of our wealth,
 opic's staff have individually pledged to donate company
 rth billions at current prices—donations that the company has
 d to matching.

le all the above private actions can be helpful, ultimately a
 omic problem this large will require government
 on. The natural policy response to an enormous economic pie
 ith high inequality (due to a lack of jobs, or poorly paid jobs,

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is progressive taxation. The tax could be general or could be against AI companies in particular. Obviously tax design is ed, and there are many ways for it to go wrong. I don't poorly designed tax policies. I think the extreme levels of predicted in this essay justify a more robust tax policy on al grounds, but I can also make a pragmatic argument to the lionaires that it's in their interest to support a good version ey don't support a good version, they'll inevitably get a bad signed by a mob.

7, I think of all of the above interventions as ways to buy time. . AI will be able to do everything, and we need to grapple It's my hope that by that time, we can use AI itself to help us e markets in ways that work for everyone, and that the ons above can get us through the transitional period.

c concentration of power

rom the problem of job displacement or economic inequality e problem of *economic concentration of power*. Section 1 the risk that humanity gets disempowered by AI, and Section d the risk that citizens get disempowered by their nts by force or coercion. But another kind of erment can occur if there is such a huge concentration of it a small group of people effectively controls government h their influence, and ordinary citizens have no influence ey lack economic leverage. Democracy is ultimately ed by the idea that the population as a whole is necessary for ion of the economy. If that economic leverage goes away, nplicit social contract of democracy may stop working. ve written about this, so I needn't go into great detail about

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t I agree with the concern, and I worry it is already starting

.

r, I am not opposed to people making a lot of money. There's
argument that it incentivizes economic growth under normal
s. I am sympathetic to concerns about impeding innovation
the golden goose that generates it. But in a scenario where
rth is 10–20% a year and AI is rapidly taking over the
yet single individuals hold appreciable fractions of the GDP,
n is *not* the thing to worry about. The thing to worry about is
wealth concentration that will break society.

famous example of extreme concentration of wealth in US
the Gilded Age, and the wealthiest industrialist of the Gilded
John D. Rockefeller. Rockefeller's wealth amounted to ~2% of
GDP at the time.⁴² A similar fraction today would lead to a
~\$600B, and the richest person in the world today (Elon
already exceeds that, at roughly \$700B. So we are already at
y unprecedented levels of wealth concentration, even *before*
the economic impact of AI. I don't think it is too much of a
we get a "country of geniuses") to imagine AI companies,
actor companies, and perhaps downstream application
s generating ~\$3T in revenue per year,⁴³ being valued at
d leading to personal fortunes well into the trillions. In that
debates we have about tax policy today simply won't apply as
in a fundamentally different situation.

this, the coupling of this economic concentration of wealth
olitical system already concerns me. AI datacenters already
a substantial fraction of US economic growth,⁴⁴ and are
gly tying together the financial interests of large tech

s (which are increasingly focused on either AI or AI culture) and the political interests of the government in a way produce perverse incentives. We already see this through the role of tech companies to criticize the US government, and the government's support for extreme anti-regulatory policies on AI.

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be done about this? First, and most obviously, companies simply choose not to be part of it. Anthropic has always strived to be a policy actor and not a political one, and to maintain our views whatever the administration. We've spoken up in favor of AI regulation and export controls that are in the public interest even when these are at odds with government policy.⁴⁵ Many have told me that we should stop doing this, that it could lead to a hostile treatment, but in the year we've been doing it, Anthropic's revenue has increased by over 6x, an almost unprecedented jump at commercial scale.

The AI industry needs a healthier relationship with government—one based on substantive policy engagement rather than mere alignment. Our choice to engage on policy substance rather than optics is sometimes read as a tactical error or failure to “read the room” rather than a principled decision, and that framing concerns me. In a healthy democracy, companies should be able to advocate for good policy on its own sake. Related to this, a public backlash against AI is possible, but this could be a corrective, but it's currently unfocused. Much of the criticism is about issues that aren't actually problems (like datacenter water usage), but it proposes solutions (like datacenter bans or poorly designed regulations) that wouldn't address the real concerns. The underlying issue that deserves attention is ensuring that AI development remains

le to the public interest, not captured by any particular
r commercial alliance, and it seems important to focus the
cussion there.

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macroeconomic interventions I described earlier in this
well as a resurgence of private philanthropy, can help to
e economic scales, addressing both the job displacement and
tion of economic power problems at once. We should look to
y of our country here: even in the Gilded Age, industrialists
Rockefeller and Carnegie felt a strong obligation to society at
ling that society had contributed enormously to their success
needed to give back. That spirit seems to be increasingly
day, and I think it is a large part of the way out of this
dilemma. Those who are at the forefront of AI's economic
uld be willing to give away both their wealth and their power.

Black seas of infinity

Effects

ection is a catchall for unknown unknowns, particularly
t could go wrong as an indirect result of positive advances in
e resulting acceleration of science and technology in general.
ve address all the risks described so far, and begin to reap the
f AI. We will likely get a “century of scientific and economic
ompressed into a decade,” and this will be hugely positive for
but we will then have to contend with the problems that
this rapid rate of progress, and those problems may come at
e may also encounter other risks that occur indirectly as a
ice of AI progress and are hard to anticipate in advance.

ure of unknown unknowns it is impossible to make an
e list, but I'll list three possible concerns as illustrative
for what we should be watching for:

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d advances in biology. If we do get a century of medical
ress in a few years, it is possible that we will greatly increase
human lifespan, and there is a chance we also gain radical
ilities like the ability to increase human intelligence or
ally modify human biology. Those would be big changes in
is possible, happening very quickly. They could be positive if
nsibly done (which is my hope, as described in *Machines of*
ig Grace), but there is always a risk they go very wrong—for
ple, if efforts to make humans smarter also make them more
ble or power-seeking. There is also the issue of “uploads” or
le brain emulation,” digital human minds instantiated in
rare, which might someday help humanity transcend its
cal limitations, but which also carry risks I find disquieting.

anges human life in an unhealthy way. A world with
ns of intelligences that are much smarter than humans at
thing is going to be a very weird world to live in. Even if AI
r't actively aim to attack humans (Section 1), and isn't
citly used for oppression or control by states (Section 3), there
t that could go wrong short of this, via normal business
itives and nominally consensual transactions. We see early
of this in the concerns about AI psychosis, AI driving people
icide, and concerns about romantic relationships with AIs. As
ample, could powerful AIs invent some new religion and
ert millions of people to it? Could most people end up
cted” in some way to AI interactions? Could people end up
; “puppeted” by AI systems, where an AI essentially watches

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every move and tells them exactly what to do and say at all, leading to a “good” life but one that lacks freedom or any of accomplishment? It would not be hard to generate dozens of these scenarios if I sat down with the creator of *Black Mirror* and tried to brainstorm them. I think this points to the importance of things like improving Claude's Constitution, over and above what is necessary for preventing the issues in Section 1. I'm sure that AI models *really* have their users' long-term interests at heart, in a way thoughtful people would endorse rather than in some subtly distorted way, seems critical.

Human purpose. This is related to the previous point, but it's not about specific human interactions with AI systems as it is about how human life changes in general in a world with powerful AI. Will humans be able to find purpose and meaning in such a world? I think this is a matter of attitude: as I said in *Machines of War*, I think human purpose does not depend on being interested in the world at something, and humans can find purpose over very long periods of time through stories and projects they love. We simply need to break the link between the creation of economic value and self-worth and meaning. But this is a transition society has to make, and there is always the risk we won't handle it well.

With all of these potential problems is that in a world with AI that we trust not to kill us, that is not the tool of an evil government, and that is genuinely working on our behalf, the AI itself to anticipate and prevent these problems. But that is guaranteed—like all of the other risks, it is something we have to take care of.

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This essay may give the impression that we are in a daunting task. I certainly found it daunting to write, in contrast with *of Loving Grace*, which felt like giving form and structure to a very beautiful music that had been echoing in my head for years. But there is much about the situation that genuinely is hard. AI threatens humanity from multiple directions, and there is a tension between the different dangers, where mitigating some risks making others worse if we do not thread the needle carefully.

We need to carefully build AI systems so they do not autonomously threaten humanity. Humanity is in genuine tension with the need for democratic control: stay ahead of authoritarian nations and not be subjugated by them. In turn, the same AI-enabled tools that are necessary to fight terrorism can, if taken too far, be turned inward to create tyranny in our countries. AI-driven terrorism could kill millions through the loss of biology, but an overreaction to this risk could lead us down to an autocratic surveillance state. The labor and economic disruption effects of AI, in addition to being grave problems in their own right, may force us to face the other problems in an environment of danger and perhaps even civil unrest, rather than being able to rely on our better angels of our nature. Above all, the sheer *number* of emerging and hiding unknown ones, and the need to deal with all of them at once, makes an intimidating gauntlet that humanity must run.

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ore, the last few years should make clear that the idea of
or even substantially slowing the technology is fundamentally
. The formula for building powerful AI systems is incredibly
much so that it can almost be said to emerge spontaneously
right combination of data and raw computation. Its creation
bly inevitable the instant humanity invented the transistor,
y even earlier when we first learned to control fire. If one
does not build it, others will do so nearly as fast. If all
s in democratic countries stopped or slowed development, by
reement or regulatory decree, then authoritarian countries
ply keep going. Given the incredible economic and military
ie technology, together with the lack of any meaningful
ent mechanism, I don't see how we could possibly convince
op.

path to a *slight* moderation in AI development that is
e with a realist view of geopolitics. That path involves
own the march of autocracies towards powerful AI for a few
enying them the resources they need to build it, ⁴⁶ namely
semiconductor manufacturing equipment. This in turn gives
c countries a buffer that they can “spend” on building
AI more carefully, with more attention to its risks, while still
g fast enough to comfortably beat the autocracies. The race
AI companies within democracies can then be handled under
lla of a common legal framework, via a mixture of industry
and regulation.

: has advocated very hard for this path, by pushing for chip
ntrols and judicious regulation of AI, but even these
common-sense proposals have largely been rejected by
ers in the United States (which is the country where it's most

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to have them). There is so much money to be made with AI—billions of dollars per year—that even the simplest measures making it difficult to overcome the political economy inherent in the trap: AI is so powerful, such a glittering prize, that it is a cult for human civilization to impose any restraints on it at all.

ine, as Sagan did in *Contact*, that this same story plays out on a scale of worlds. A species gains sentience, learns to use tools, and faces the exponential ascent of technology, faces the crises of environmental degradation and nuclear weapons, and if it survives those, it faces the hardest and final challenge when it learns how to shape machines that think. Whether we survive that test and go on to the beautiful society described in *Machines of Loving Grace*, or to slavery and destruction, will depend on our character and our civilization as a species, our spirit and our soul.

Despite many obstacles, I believe humanity has the strength inside to pass this test. I am encouraged and inspired by the thousands of researchers who have devoted their careers to helping us understand and improve AI models, and to shaping the character and constitution of the world. I think there is now a good chance that those efforts bear fruit. I am encouraged that at least some companies are beginning to matter. I am encouraged that they'll pay meaningful commercial costs to block their own contribution to the threat of bioterrorism. I am encouraged that a few brave people have resisted the prevailing winds and passed legislation that puts the first early seeds of guardrails on AI systems. I am encouraged that the public demands that AI carries risks and wants those risks addressed. I am encouraged by the indomitable spirit of freedom around the world and the determination to resist tyranny wherever it occurs.

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ll need to step up our efforts if we want to succeed. The first those closest to the technology to simply tell the truth about on humanity is in, which I have always tried to do; I'm doing explicitly and with greater urgency with this essay. The next e convincing the world's thinkers, policymakers, companies, as of the imminence and overriding importance of this issue worth expending thought and political capital on this in n to the thousands of other issues that dominate the news

Then there will be a time for courage, for enough people to prevailing trends and stand on principle, even in the face of their economic interests and personal safety.

in front of us will be impossibly hard, asking more of us than ve can give. But in my time as a researcher, leader, and citizen, n enough courage and nobility to believe that we can win— put in the darkest circumstances, humanity has a way of seemingly at the last minute, the strength and wisdom prevail. We have no time to lose.



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ymmetric to a point I made in *Machines of Loving Grace*,
arted by saying that AI's upsides shouldn't be thought of in
prophecy of salvation, and that it's important to be concrete
ded and to avoid grandiosity. Ultimately, prophecies of
and prophecies of doom are unhelpful for confronting the
, for basically the same reasons. ↩

ic's goal is to remain consistent through such changes. When
out AI risks was politically popular, Anthropic cautiously
for a judicious and evidence-based approach to these risks.
talking about AI risks is politically unpopular, Anthropic
to cautiously advocate for a judicious and evidence-based
to these risks. ↩

ie, I have gained increasing confidence in the trajectory of AI
elihood that it will surpass human ability across the board,
uncertainty still remains. ↩

ontrols for chips are a great example of this. They are simple
r to mostly just work. ↩

ourse, the hunt for such evidence must be intellectually
ch that it could also turn up evidence of a lack of danger.
ncy through model cards and other disclosures is an attempt
intellectually honest endeavor. ↩

since writing *Machines of Loving Grace* in 2024, AI systems
me capable of doing tasks that take humans several hours,
I recently assessing that Opus 4.5 can do about four human
work with 50% reliability. ↩

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re clear, even if powerful AI is only 1–2 years away in a sense, many of its societal consequences, both positive and may take a few years longer to occur. This is why I can ously think that AI will disrupt 50% of *entry-level* white-over 1–5 years, while also thinking we may have AI that is ble than *everyone* in only 1–2 years. ↩

th adding that the *public* (as compared to policymakers) does e very concerned with AI risks. I think some of their focus is e. AI job displacement), and some is misguided (such as about water use of AI, which is not significant). This backlash hope that a consensus around addressing risks is possible, but is not yet been translated into policy changes, let alone r well-targeted policy changes. ↩

1 also, of course, manipulate (or simply pay) large numbers of ito doing what they want in the physical world. ↩

think this is a straw man: it's my understanding, for example, LeCun holds this position. ↩

mple, see Section 5.5.2 (p. 63–66) of the Claude 4 .d. ↩

re also a number of other assumptions inherent in the simple ich I won't discuss here. Broadly, they should make us less ount the specific simple story of misaligned power-seeking, ore worried about possible unpredictable behavior we ticipated. ↩

Game describes a version of this involving humans rather

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mple, models may be told not to do various bad things, and by humans, but may then observe that many humans do those bad things! It's not clear how this contradiction would aid a well-designed constitution should encourage the model these contradictions gracefully), but this type of dilemma is inherent from the supposedly "artificial" situations that we put in during testing. ↩

tally, one consequence of the constitution being a natural-document is that it is legible to the world, and that means it is queried by anyone and compared to similar documents by companies. It would be valuable to create a race to the top that encourages companies to release these documents, but asks them to be good. ↩

even a hypothesis about a deep unifying principle bridging the character-based approach from Constitutional AI to multimodal interpretability and alignment science. According to the research, the fundamental mechanisms driving Claude originally allow for it to simulate characters in pretraining, such as generating what the characters in a novel would say. This would suggest a useful way to think about the constitution is more like a description that the model uses to instantiate a consistent character—it would also help us explain the "I must be a bad person" mentioned above (because the model is trying to *act as if* it's a character—in this case a bad one), and would suggest that interpretability methods should be able to discover "psychological" patterns within models. Our researchers are working on ways to test this hypothesis. ↩

clear, monitoring is done in a privacy-preserving way. ↩

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our own experiments with what are essentially voluntarily
ules with our Responsible Scaling Policy, we have found over
gain that it's very easy to end up being too rigid, by drawing
seem important ex ante but turn out to be silly in retrospect.
ry easy to set rules about the wrong things when a
y is advancing rapidly. ↩

and RAISE do not apply at all to companies with under
annual revenue. They only apply to larger, more established
s like Anthropic. ↩

ally read Joy's essay 25 years ago, when it was written, and it
ound impact on me. Then and now, I do see it as too
c—I don't think broad "relinquishment" of whole areas of
y, which Joy suggests, is the answer—but the issues it raises
risingly prescient, and Joy also writes with a deep sense of
n and humanity that I admire. ↩

ave to worry about state actors, now and in the future, and I
at in the next section. ↩

s evidence that many terrorists are at least relatively well-
which might seem to contradict what I'm arguing here about
correlation between ability and motivation. But I think in
they are compatible observations: if the ability threshold for
al attack is high, then almost by definition those who *currently*
ust have high ability, even if ability and motivation are
correlated. But in a world where the limitations on ability
oved (e.g., with future LLMs), I'd predict that a substantial
n of people with the motivation to kill but lower ability
t to do so—just as we see for crimes that don't require much
e school shootings). ↩

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Shinrikyo did try, however. The leader of Aum Shinrikyo, do, had training in virology from Kyoto University, and to produce both anthrax and ebola. However, as of n he lacked enough expertise and resources to succeed at ar is now substantially lower, and LLMs could reduce it er. ↩

re phenomenon relating to mass murderers is that the style of ey choose operates almost as a grotesque sort of fad. In the l 1980s, serial killers were very common, and new serial en copied the behavior of more established or famous serial the 1990s and 2000s, mass shootings became more common, al killers became less common. There is no technological at triggered these patterns of behavior, it just appears that rderers were copying each others' behavior and the thing to copy changed. ↩

jailbreakers sometimes believe that they've compromised ifiers when they get the model to output one specific piece of on, such as the genome sequence of a virus. But as I explained e threat model we are worried about involves step-by-step, e advice that extends over weeks or months about specific eps in the bioweapons production process, and this is what iers aim to defend against. (We often describe our s looking for “universal” jailbreaks—ones that don't just re specific or narrow context, but broadly open up the havior.) ↩

we will continue to invest in work to make our classifiers ient, and it may make sense for companies to share advances with one another. ↩

sly, I do not think companies should have to disclose details about the specific steps in biological weapons that they are blocking, and the transparency legislation that passed so far (SB 53 and RAISE) accounts for this issue. ↩

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r related idea is “resilience markets” where the government stockpiling of PPE, respirators, and other essential needed to respond to a biological attack by promising time to pay a pre-agreed price for this equipment in an y. This incentivizes suppliers to stockpile such equipment ar that the government will seize it without compensation. ↩

n I more worried about large actors for seizing power, but rs for causing destruction? Because the dynamics are Seizing power is about whether one actor can amass enough o overcome everyone else—thus we should worry about the erful actors and/or those closest to AI. Destruction, by an be wrought by those with little power if it is much harder against than to cause. It is then a game of defending against *numerous* threats, which are likely to be smaller actors. ↩

ght sound like it is in tension with my point that attack and ay be more balanced with cyberattacks than with ns, but my worry here is that if a country's AI is the most n the world, then others will not be able to defend even if the y itself has an intrinsic attack-defense balance. ↩

mple, in the United States this includes the fourth nt and the Posse Comitatus Act. ↩

be clear, there are some arguments for building large :s in countries with varying governance structures,

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ly if they are controlled by companies in democracies. Such
could in principle help democracies compete better with the
h is the greater threat. I also think such datacenters don't
a risk unless they are very large. But on balance, I think
warranted when placing very large datacenters in countries
stitutional safeguards and rule-of-law protections are less
lished. ↩

of course, also an argument for improving the security of the
deterrent to make it more likely to be robust against powerful
uclear-armed democracies should do this. But we don't know
powerful AI will be capable of or which defenses, if any, will
st it, so we should not assume that these measures will
y solve the problem. ↩

s also the risk that even if the nuclear deterrent remains
in attacking country might decide to call our bluff—it's
ether we'd be willing to use nuclear weapons to defend
lone swarm even if the drone swarm has a substantial risk of
g us. Drone swarms might be a new thing that is less severe
ar attacks but more severe than conventional attacks.
ely, differing assessments of the effectiveness of the nuclear
in the age of AI might alter the game theory of nuclear
a destabilizing manner. ↩

lear, I would believe it is the right strategy not to sell chips to
en if the timeline to powerful AI were substantially longer. We
the Chinese “addicted” to American chips—they are
d to develop their native chip industry one way or another. It
hem many years to do so, and all we are doing by selling them
ving them a big boost during that time. ↩

lear, most of what is being used in Ukraine and Taiwan
not *fully* autonomous weapons. These are coming, but not
7. ↩

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odel card for Claude Opus 4.5, our most recent model, shows
performs better on a performance engineering interview
given at Anthropic than any interviewee in the history of
my. ↩

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g all of the code” and “doing the task of a software engineer
l” are very different things, because software engineers do
e than just write code, including testing, dealing with
ents, files, and installation, managing cloud compute
nts, iterating on products, and much more. ↩

4. Player
piano

ters are general in a sense, but are clearly incapable on their
e vast majority of human cognitive abilities, even as they
ceed humans in a few areas (such as arithmetic). Of course,
It *on top* of computers, such as AI, are now capable of a wide
ognitive abilities, which is what this essay is about. ↩

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lear, AI models do not have precisely the same profile of
and weaknesses as humans. But they are also advancing fairly
along every dimension, such that having a spiky or uneven
y not ultimately matter. ↩

1 there is debate among economists about this idea. ↩

al wealth is a “stock,” while GDP is a “flow,” so this isn't a claim
efeller owned 2% of the economic value in the United States.
rder to measure the total wealth of a nation than the GDP,
e's individual incomes vary a lot per year, so it's hard to make

he same units. The ratio of the largest personal fortune to the not comparing apples to apples, is nevertheless a perfectly benchmark for extreme wealth concentration. ↩

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al value of labor across the economy is \$60T/year, so would correspond to 5% of this. That amount could be a company that supplied labor for 20% of the cost of and had 25% market share, even if the demand for labor did d (which it almost certainly would due to the lower cost). ↩

lear, I do not think actual AI productivity is yet responsible antial fraction of US economic growth. Rather, I think the spending represents growth caused by anticipatory it that amounts to the market expecting *future* AI-driven growth and investing accordingly. ↩

ve agree with the administration, we say so, and we look for agreement where mutually supported policies are genuinely he world. We are aiming to be honest brokers rather than opponents of any given political party. ↩

think anything more than a few years is possible: on longer , they will build their own chips. ↩

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